Derivatives of the Variational Energy

Gradient and Hessian formulas used in Variational Monte Carlo

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Setup and notation

Let $\psi_{\boldsymbol{\alpha}}(x)$ be a (real-valued) variational wavefunction depending on parameters $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_P)$ and x denote many-body coordinates.

Variational energy (Rayleigh quotient)

$$E(\boldsymbol{\alpha}) = \frac{\langle \psi_{\boldsymbol{\alpha}} | H | \psi_{\boldsymbol{\alpha}} \rangle}{\langle \psi_{\boldsymbol{\alpha}} | \psi_{\boldsymbol{\alpha}} \rangle} = \frac{\int dx\, \psi_{\boldsymbol{\alpha}}(x)\, [H\psi_{\boldsymbol{\alpha}}](x)}{\int dx\, \psi_{\boldsymbol{\alpha}}(x)^2}.$$

Define normalization $\mathcal{N}(\boldsymbol{\alpha}) := \int dx\, \psi_{\boldsymbol{\alpha}}(x)^2$.

Local energy and sampling distribution

Local energy:

$$E_L(x;\boldsymbol{\alpha}) := \frac{[H\psi_{\boldsymbol{\alpha}}](x)}{\psi_{\boldsymbol{\alpha}}(x)}.$$

Sampling distribution:

$$p_{\boldsymbol{\alpha}}(x) := \frac{\psi_{\boldsymbol{\alpha}}(x)^2}{\mathcal{N}(\boldsymbol{\alpha})}, \qquad \int dx\, p_{\boldsymbol{\alpha}}(x) = 1.$$

Then

$$E(\boldsymbol{\alpha}) = \int dx\, p_{\boldsymbol{\alpha}}(x)\, E_L(x;\boldsymbol{\alpha}) = \langle E_L \rangle\,.$$

Differentiate the Rayleigh quotient

Write

$$E = \frac{A}{B}, \qquad A = \int dx\, \psi_{\boldsymbol{\alpha}}\, H\psi_{\boldsymbol{\alpha}}, \quad B = \int dx\, \psi_{\boldsymbol{\alpha}}^2.$$

Then

$$\partial_\alpha E = \frac{(\partial_\alpha A)B - A(\partial_\alpha B)}{B^2}.$$

Compute

$$\partial_\alpha B = \int dx\, 2\psi_{\boldsymbol{\alpha}}\, \partial_\alpha \psi_{\boldsymbol{\alpha}},$$

and using Hermiticity of H,

$$\partial_\alpha A = \int dx \left[(\partial_\alpha \psi_{\boldsymbol{\alpha}})(H\psi_{\boldsymbol{\alpha}}) + \psi_{\boldsymbol{\alpha}}\, H(\partial_\alpha \psi_{\boldsymbol{\alpha}})\right] = 2\int dx\, (\partial_\alpha \psi_{\boldsymbol{\alpha}})\, H\psi_{\boldsymbol{\alpha}}.$$

Gradient in expectation-value form

Insert into the quotient rule and use $A = EB$:

$$\partial_\alpha E = 2\left[\frac{\int dx\,(\partial_\alpha \psi_{\boldsymbol{\alpha}})\,H\psi_{\boldsymbol{\alpha}}}{\int dx\,\psi_{\boldsymbol{\alpha}}^2} - E\,\frac{\int dx\,\psi_{\boldsymbol{\alpha}}\,\partial_\alpha \psi_{\boldsymbol{\alpha}}}{\int dx\,\psi_{\boldsymbol{\alpha}}^2}\right].$$

Rewrite with $E_L = (H\psi_{\boldsymbol{\alpha}})/\psi_{\boldsymbol{\alpha}}$:

$$\frac{\int dx\,(\partial_\alpha \psi_{\boldsymbol{\alpha}})\,H\psi_{\boldsymbol{\alpha}}}{\int dx\,\psi_{\boldsymbol{\alpha}}^2} = \frac{\int dx\,\psi_{\boldsymbol{\alpha}}^2\left(\frac{\partial_\alpha \psi_{\boldsymbol{\alpha}}}{\psi_{\boldsymbol{\alpha}}}\right)E_L}{\int dx\,\psi_{\boldsymbol{\alpha}}^2} = \left\langle\left(\frac{\partial_\alpha \psi_{\boldsymbol{\alpha}}}{\psi_{\boldsymbol{\alpha}}}\right)E_L\right\rangle.$$

Thus

$$\boxed{\partial_\alpha E = 2\left(\left\langle\left(\frac{\partial_\alpha \psi_{\boldsymbol{\alpha}}}{\psi_{\boldsymbol{\alpha}}}\right)E_L\right\rangle - \left\langle\frac{\partial_\alpha \psi_{\boldsymbol{\alpha}}}{\psi_{\boldsymbol{\alpha}}}\right\rangle\langle E_L\rangle\right).}$$

Log-derivative identity (important simplification)

A key algebraic identity is

$$\boxed{\frac{\partial_\alpha \psi_{\boldsymbol{\alpha}}(x)}{\psi_{\boldsymbol{\alpha}}(x)} = \partial_\alpha \ln \psi_{\boldsymbol{\alpha}}(x).}$$

Definition (log-derivative observable)

$$O_\alpha(x) := \partial_\alpha \ln \psi_{\boldsymbol{\alpha}}(x).$$

Then the gradient becomes the covariance form

$$\boxed{\partial_\alpha E = 2\left(\langle O_\alpha E_L\rangle - \langle O_\alpha\rangle \langle E_L\rangle\right) = 2\,\mathrm{Cov}(O_\alpha, E_L).}$$

Complex case (brief remark): if ψ is complex, define $O_\alpha = \partial_\alpha \ln \psi$ and use

$$\partial_\alpha E = 2\ \mathrm{Re}(\langle O_\alpha^* E_L\rangle - \langle O_\alpha^*\rangle \langle E_L\rangle)\,.$$

Key identity: derivative of an expectation value

Let

$$\langle A \rangle = \int dx\, p_{\boldsymbol{\alpha}}(x)\, A(x;\boldsymbol{\alpha}), \qquad p_{\boldsymbol{\alpha}}(x) = \frac{\psi_{\boldsymbol{\alpha}}(x)^2}{\mathcal{N}}.$$

Then for real $\psi_{\boldsymbol{\alpha}}$ one can show

$$\boxed{\partial_\beta \langle A \rangle = \langle \partial_\beta A \rangle + 2\left(\langle A\, O_\beta \rangle - \langle A \rangle \langle O_\beta \rangle\right),} \qquad O_\beta = \partial_\beta \ln \psi_{\boldsymbol{\alpha}}.$$

- The first term accounts for explicit $\boldsymbol{\alpha}$-dependence of A.
- The second term accounts for how the sampling distribution changes with $\boldsymbol{\alpha}$.

Useful derivative: $\partial_\beta E_L$

Since $E_L = (H\psi_{\boldsymbol{\alpha}})/\psi_{\boldsymbol{\alpha}}$, its parameter derivative is

$$\partial_\beta E_L = \partial_\beta \left(\frac{H\psi_{\boldsymbol{\alpha}}}{\psi_{\boldsymbol{\alpha}}} \right) = \frac{H(\partial_\beta \psi_{\boldsymbol{\alpha}})}{\psi_{\boldsymbol{\alpha}}} - E_L \frac{\partial_\beta \psi_{\boldsymbol{\alpha}}}{\psi_{\boldsymbol{\alpha}}}.$$

Using $O_\beta = \partial_\beta \ln \psi_{\boldsymbol{\alpha}} = (\partial_\beta \psi_{\boldsymbol{\alpha}})/\psi_{\boldsymbol{\alpha}}$:

$$\boxed{\partial_\beta E_L = \frac{H(\partial_\beta \psi_{\boldsymbol{\alpha}})}{\psi_{\boldsymbol{\alpha}}} - E_L\, O_\beta.}$$

Notation

Also define the second log-derivative

$$\boxed{O_{\alpha\beta}(x) := \partial_\beta O_\alpha(x) = \partial_\alpha \partial_\beta \ln \psi_{\boldsymbol{\alpha}}(x).}$$

Hessian: start from the covariance gradient

We have

$$\partial_\alpha E = 2\left(\langle O_\alpha E_L\rangle - \langle O_\alpha\rangle\langle E_L\rangle\right).$$

Differentiate w.r.t. β:

$$\partial_\beta\partial_\alpha E = 2\left[\partial_\beta\langle O_\alpha E_L\rangle - \partial_\beta(\langle O_\alpha\rangle\langle E_L\rangle)\right].$$

We now apply the expectation-derivative identity to each expectation value.

Explicit Hessian formula (practical VMC form)

Using $\partial_\beta \langle A \rangle = \langle \partial_\beta A \rangle + 2(\langle A O_\beta \rangle - \langle A \rangle \langle O_\beta \rangle)$, with $A = O_\alpha E_L$, $A = O_\alpha$, and $A = E_L$, we obtain:

$$\boxed{\begin{aligned}\partial_\beta \partial_\alpha E = 2\Big[& \langle O_{\alpha\beta}\, E_L \rangle + \langle O_\alpha\, \partial_\beta E_L \rangle - \langle O_{\alpha\beta} \rangle \langle E_L \rangle - \langle O_\alpha \rangle \langle \partial_\beta E_L \rangle \\ & + 2\Big(\langle O_\alpha\, E_L\, O_\beta \rangle - \langle O_\alpha\, E_L \rangle \langle O_\beta \rangle - \langle O_\alpha\, O_\beta \rangle \langle E_L \rangle - \langle O_\alpha \rangle \langle E_L\, O_\beta \rangle + 2 \langle O_\alpha \rangle \langle O_\beta \rangle \langle \end{aligned}}$$

- First line: "explicit" derivatives ($O_{\alpha\beta}$ and $\partial_\beta E_L$).
- Second line: "distribution-derivative" terms (a third-order cumulant-like combination).

Compact cumulant interpretation (optional)

Define centered variables

$$\delta O_\alpha = O_\alpha - \langle O_\alpha \rangle \,, \quad \delta O_\beta = O_\beta - \langle O_\beta \rangle \,, \quad \delta E_L = E_L - \langle E_L \rangle \,.$$

Then the gradient is

$$\partial_\alpha E = 2 \, \langle \delta O_\alpha \, \delta E_L \rangle \,.$$

The Hessian can be viewed as

$$\partial_\beta \partial_\alpha E = 2 \, \mathrm{Cov}(O_{\alpha\beta}, E_L) + 2 \, \mathrm{Cov}(O_\alpha, \partial_\beta E_L) + 4 \, \kappa_3(O_\alpha, O_\beta, E_L),$$

where κ_3 denotes the third cumulant (connected three-point function)

$$\kappa_3(O_\alpha, O_\beta, E_L) = \langle \delta O_\alpha \, \delta O_\beta \, \delta E_L \rangle \,.$$

This form highlights why Hessian estimates are typically noisier than gradients in Monte Carlo.

Monte Carlo estimators (implementation hint)

Given samples $\{x^{(m)}\}_{m=1}^{M} \sim p_{\boldsymbol{\alpha}}$, compute

$$O_\alpha^{(m)} = \partial_\alpha \ln \psi_{\boldsymbol{\alpha}}(x^{(m)}), \quad O_\beta^{(m)} = \partial_\beta \ln \psi_{\boldsymbol{\alpha}}(x^{(m)}), \quad E_L^{(m)} = E_L(x^{(m)}),$$

and, if needed,

$$O_{\alpha\beta}^{(m)} = \partial_\alpha \partial_\beta \ln \psi_{\boldsymbol{\alpha}}(x^{(m)}), \qquad (\partial_\beta E_L)^{(m)} = \partial_\beta E_L\big|_{x^{(m)}}.$$

Practical remark

Many VMC optimizers avoid explicit Hessians (noisy) and instead use stochastic reconfiguration / natural-gradient methods that only require first derivatives.

Summary

- Define $p(x) \propto \psi(x)^2$ and $E_L = (H\psi)/\psi$ so $E = \langle E_L \rangle$.
- Log-derivative identity:
$$\frac{\partial_\alpha \psi}{\psi} = \partial_\alpha \ln \psi.$$
- Gradient:
$$\partial_\alpha E = 2\left(\langle O_\alpha E_L \rangle - \langle O_\alpha \rangle \langle E_L \rangle\right) = 2\,\mathrm{Cov}(O_\alpha, E_L).$$
- Hessian: obtained by differentiating the covariance and accounting for changes in $p(x)$, yielding an explicit VMC-ready expression involving $O_{\alpha\beta}$ and $\partial_\beta E_L$.